

Week 2 Update — COVID-19 Global Data Analysis

Shrey Jain • Analyzing global case & death trends over time • Prepared for Neelam Gupta

Weekly update

This week I analyzed how global COVID-19 cases and deaths evolved over time using the OWID *World* and continent series. I built daily (7-day smoothed) case and death trends, cumulative totals, a case-fatality-ratio (CFR) trend, and a by-continent death breakdown, all as reproducible Plotly charts in a notebook.

Discovered:

- Cases and deaths **decoupled** after early 2021: peak deaths (~14.8K/day, 24 Jan 2021) came before the largest case wave (~6.3M/day, 25 Dec 2022), which was far milder in deaths.
- The naive cumulative **CFR fell from ~8.0% (Apr 2020) to ~0.9%**, driven mainly by expanded testing plus later reductions in severity (vaccination, variants).
- By year, **2021 was deadliest** (~3.5M deaths from ~200M cases) while **2022 had the most cases** (~424M) with far fewer deaths (~1.25M).

Next: break waves down by variant period, compare deaths against vaccination coverage, and bring in excess-mortality where available as a more robust mortality measure.

Analysis: global case & death trends

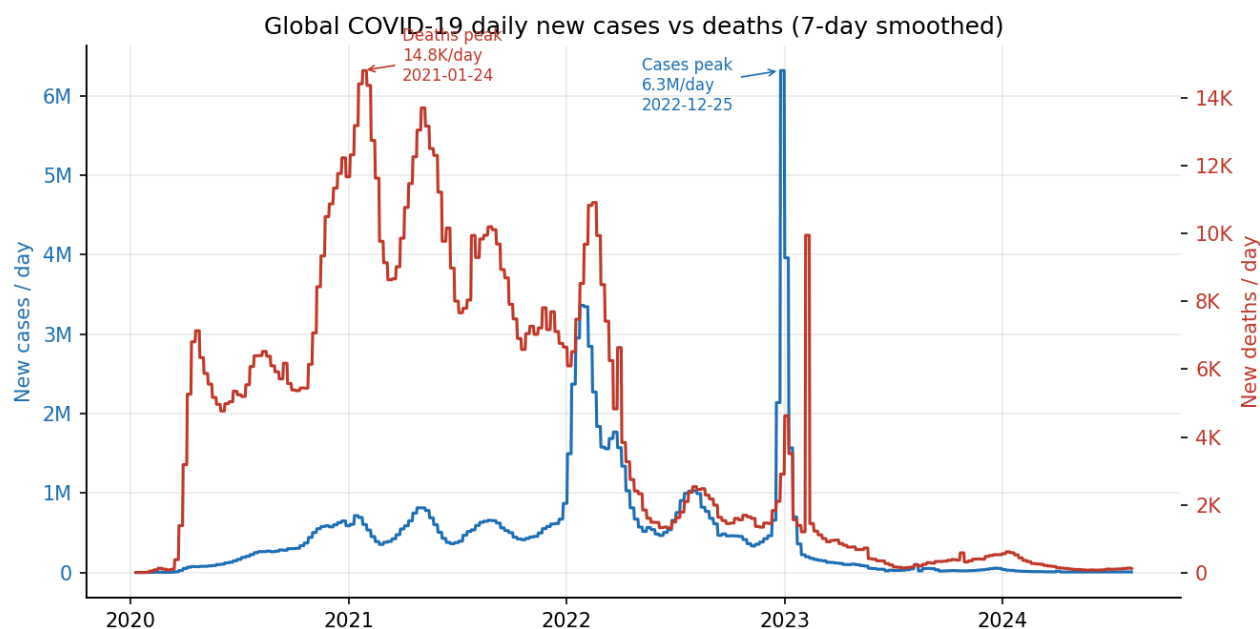


Fig 1. Daily new cases (blue) vs deaths (red), 7-day smoothed. The peaks occur at different times — the clearest sign of case–death decoupling.

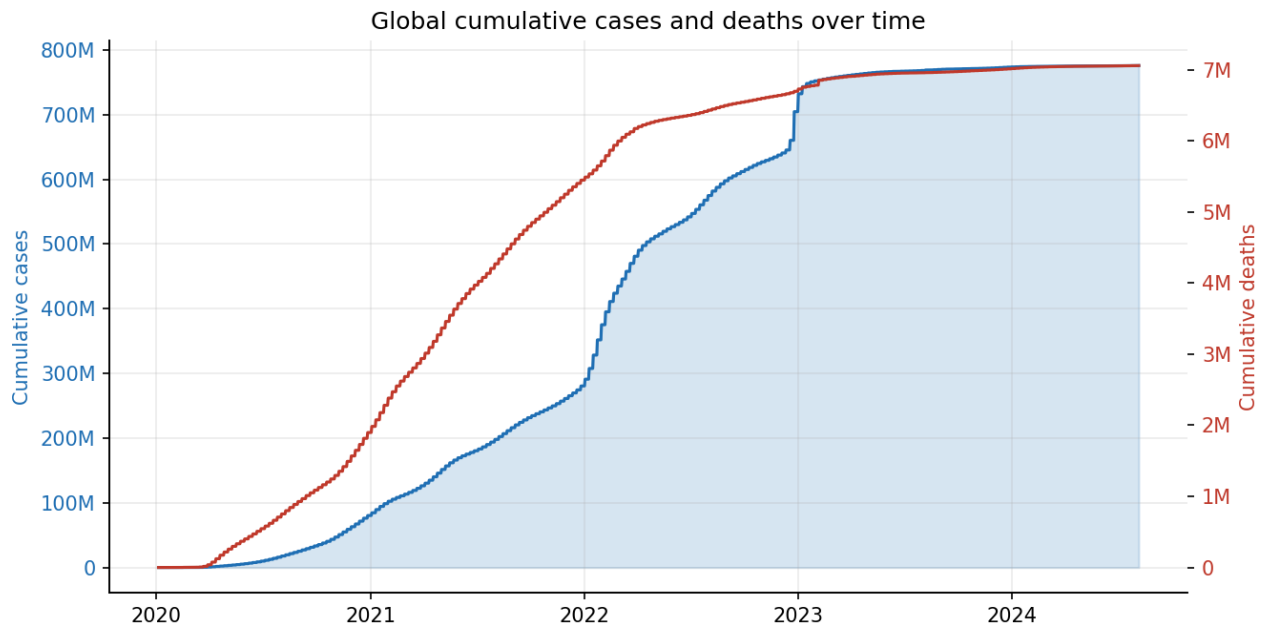


Fig 2. Cumulative cases and deaths. Both plateau after 2023 as reporting wound down; final totals ~776M cases and ~7.06M deaths.

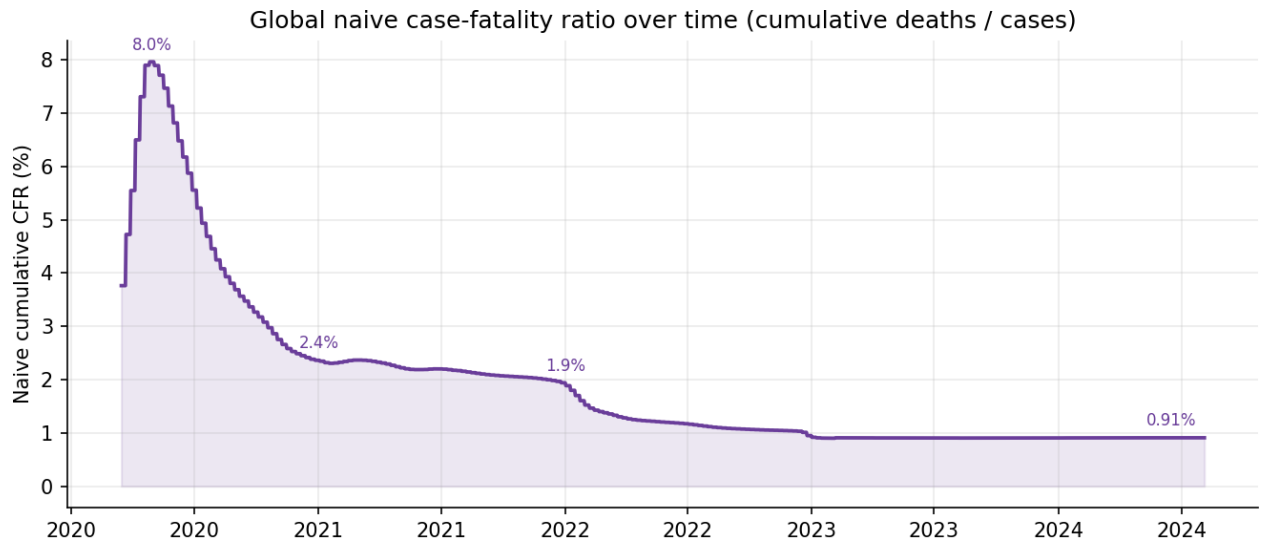


Fig 3. Naive cumulative CFR (deaths / cases). The steep early value reflects limited testing; it settles below 1% as testing scaled up.

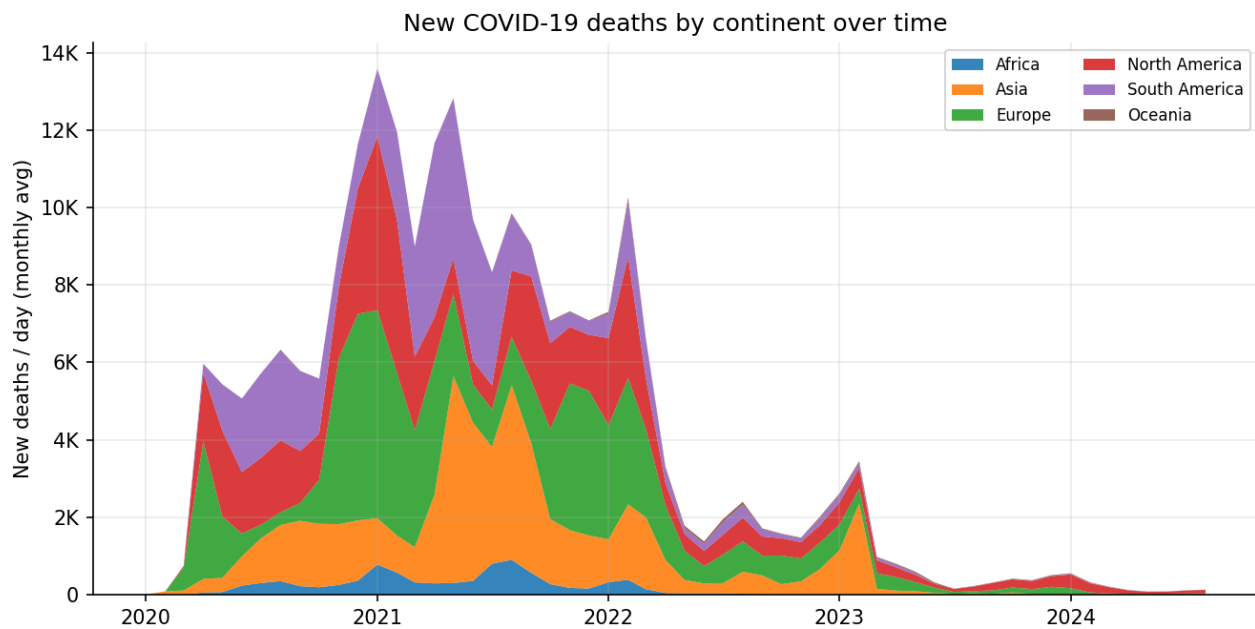


Fig 4. New deaths by continent (monthly average). Europe and the Americas dominate the early waves; Asia contributes large later waves.

Work artifacts & data sources

Repository	github.com/shreyjain99/covid19-global-analysis
Week 2 notebook	notebooks/week2_case_death_trends.ipynb (interactive Plotly)
Charts	outputs/week2/fig1-fig4 (.png)
Data source	Our World in Data COVID-19 dataset (github.com/owid/covid-19-data)
Data coverage	429,435 rows x 67 cols; 2020-01-01 to 2024-08-14; ~230 countries

Reproducibility (how to re-run)

- `git clone` the repo and `pip install -r requirements.txt`.
- `python scripts/download_data.py` to fetch the dataset into `data/raw/`.
- Open `notebooks/week2_case_death_trends.ipynb` and Run All — every figure regenerates from the raw data with no manual steps.

Risks / blockers & support needed

- **No blockers to delivery.** The week's objective is met and reproducible.
- **Data caveat:** naive CFR is biased by testing coverage and reporting lags, so it understates true lethality early and isn't comparable across countries without adjustment.
- **Support:** confirm whether excess-mortality (sparse here, ~3% populated) should be sourced separately for a more robust mortality view in a later week.